HW4

Tuesday, 7 February 2023 9:19 AM

Question 10.2

$$\Phi^2 = -\frac{R^2 D^{-1} K}{A} = -\phi^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1000 \end{bmatrix}^{-1} \begin{bmatrix} -2 & 1 & 0 \\ 2 & -2 & 2 \\ 0 & 1 & -2 \end{bmatrix}$$

$N' = \Phi^{-1} \tanh \Phi$ → show in terms of ϕ

Evaluating Φ^2 expression: (using wolfram Alpha)

$$\Phi^2 = -\phi^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0.001 \end{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ 2 & -2 & 2 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\Phi^2 = -\phi^2 \begin{bmatrix} -2 & 1 & 0 \\ +2 & -2 & +2 \\ 0 & 0.001 & -0.002 \end{bmatrix}$$

From notes:

$$\Phi^{-1} \tanh \Phi = P_0 + \sum_{j=1}^{n-1} \lambda_j^{-1} \tanh(\lambda_j P_j)$$; P_j → projection matrix

$$P_j = \frac{u_j v_j^T}{v_j^T u_j}$$ u → right eigenvector; v → left eigenvector

P_0 → projection matrix corresponding to $\lambda = 0$

Find $\Phi = \sqrt{\Phi^2}$ from python:

$$\Phi = \begin{bmatrix} 1.31 & -0.383 & -0.54 \\ -0.766 & 1.306 & -1.845 \\ -5.4\times10^{-4} & -9.23\times10^{-4} & 2.386\times10^{-3} \end{bmatrix}$$ → eigenvalues $\lambda_1 = 1.8478$, $\lambda_2 = 0.7665$, $\lambda_3 = 0$

Finding left and rite eigenvalues to construct the projection matrix:

$$N' = \Phi^{-1} \tanh \Phi = \begin{bmatrix} 0 & 0 & 0.997 \\ 0 & 0 & 1.994 \\ 0 & 0 & 0.997 \end{bmatrix} + \frac{\tanh(\lambda_1 \phi)}{\lambda_1} \begin{bmatrix} 0.499 & -0.35 & 0.207 \\ -0.707 & 0.5 & -0.293 \\ 0 & 0 & 0 \end{bmatrix} + \frac{\tan(\lambda_2 \phi)}{\lambda_2} \begin{bmatrix} 0.499 & 0.352 & -1.204 \\ 0.705 & 0.497 & -1.701 \\ 0 & 0 & 0 \end{bmatrix}$$

$\lambda_1 = 1.85$ $\lambda_2 = 0.767$

(b) Selectivity to paraxylene; selectivity = $\frac{\text{desired product}}{\text{undesired product}}$

From class,

reaction rate = $\hat{K} N' C_0$; $C_0 = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}$ (Conc. vector at pore mouth) ; $\hat{K}$ → rate constant matrix per unit area

(for every C_0)

r_3 = reaction rate of paraxylene described by [Third row of $kN'C_0$ matrix]

r_2 = reaction rate of Metaxylene [Second row of $kN'C_0$ matrix]

r_1 = reaction rate of Ortho [First row of $kN'C_0$ matrix]

$$\text{Selectivity} = \frac{r_3}{r_2 + r_1} = \frac{e_3^T \hat{K} N' c_0}{e_1^T \hat{K} N' c_0 + e_2^T \hat{K} N' c_0}$$

e vectors pick out the specific rows defining the rates

(c) Code solves for the various conditions → Attached

Analysis:

It seems the diffusion coefficient for para (A_3) is order of magnitude greater than those of ortho and meta. N' has inverse diffusion dependence but it is nested in the inverse and square roots

→ Selectivities of -1 means that any para formed is being consumed again to form different isomers.

Question 2 : GA